

```

clc;
clear;
close all;

Rp = 1;
Rs = 40;
Wp = 0.5;

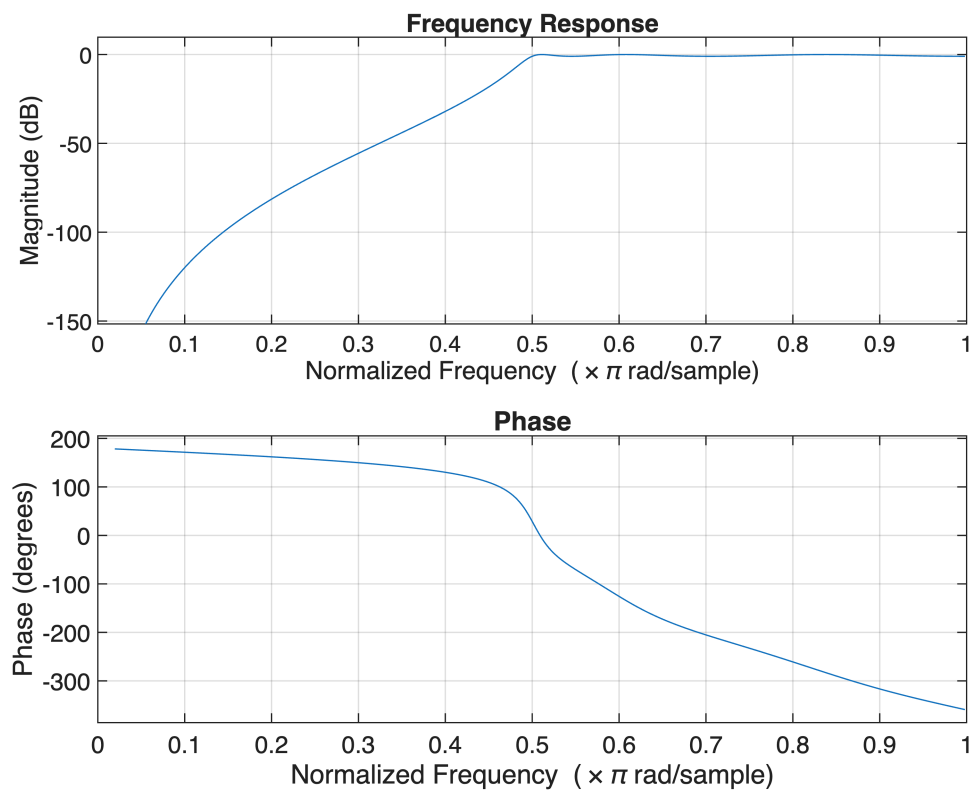
Ws = 0.35;

[n,Wn] = cheb1ord(Wp,Ws,Rp,Rs);

[b,a] = cheby1(n,Rp,Wn,'high');

figure;
freqz(b,a);
title('Frequency Response');

```

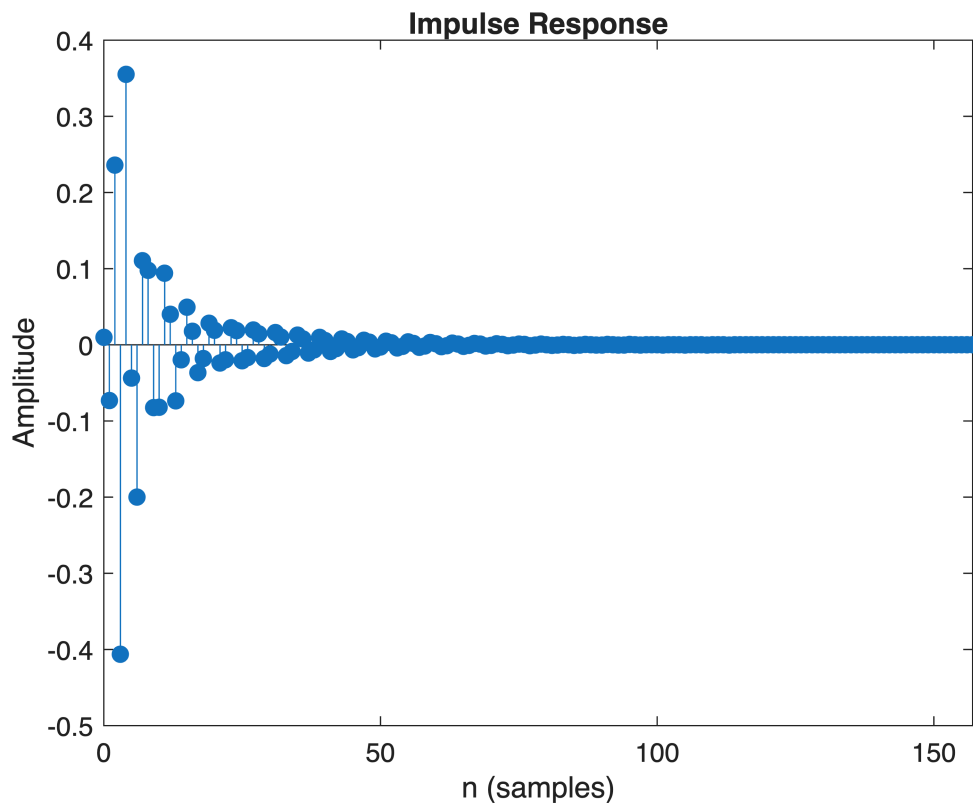


```

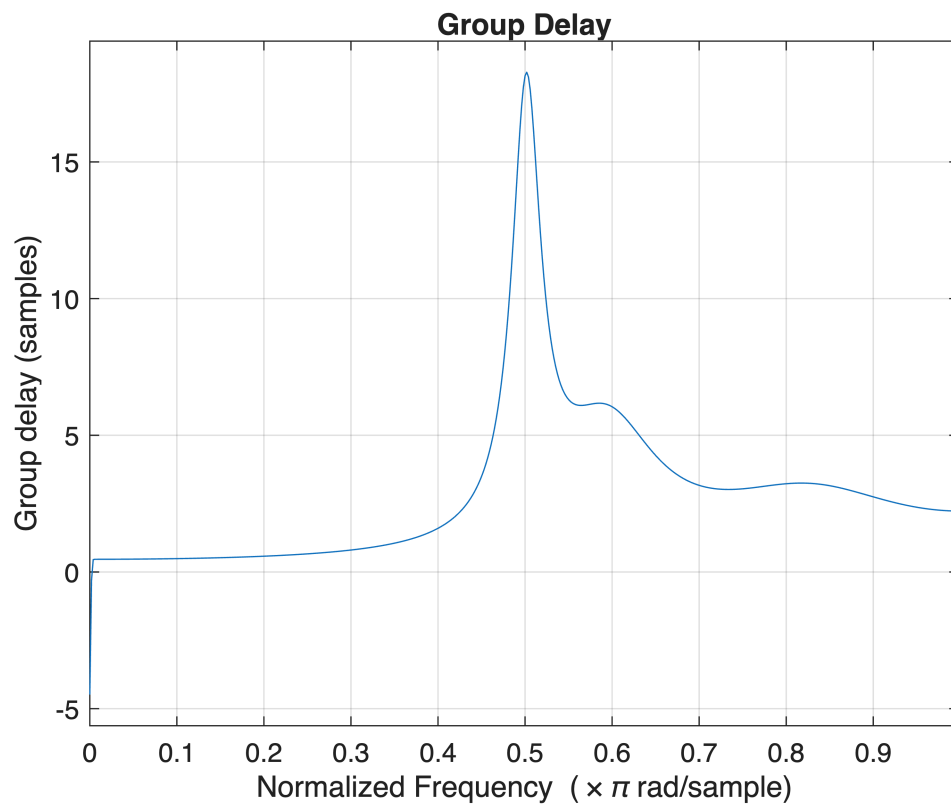
figure;

```

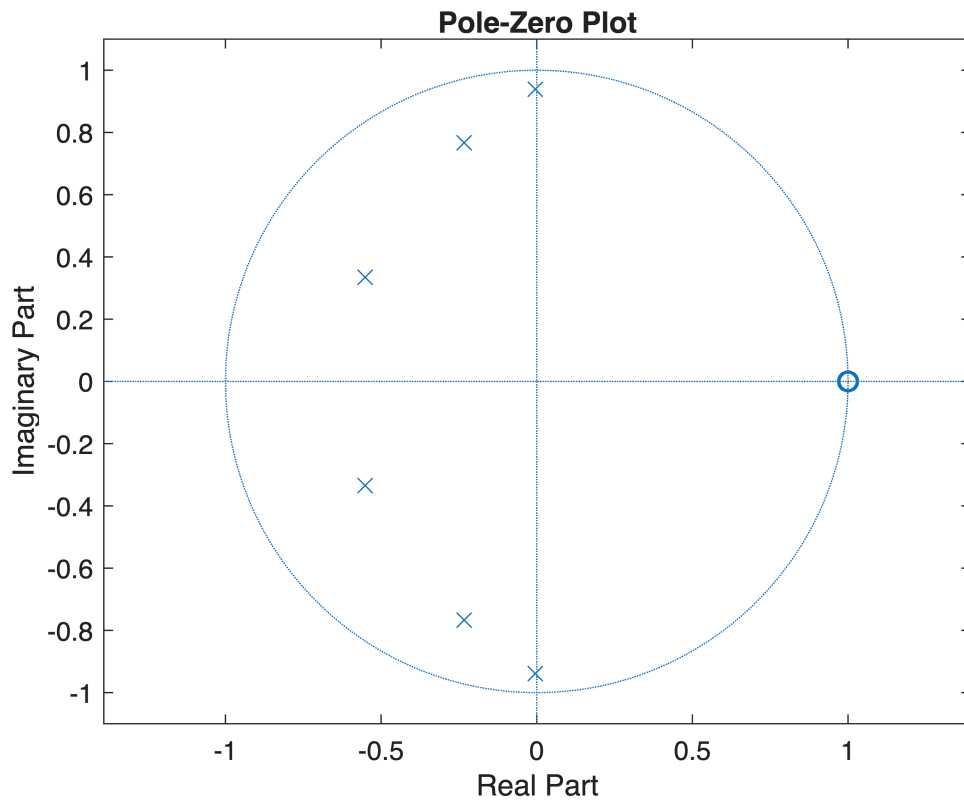
```
impz(b,a);  
title('Impulse Response');
```



```
figure;  
grpdelay(b,a);  
title('Group Delay');
```



```
figure;  
zplane(b,a);  
title('Pole-Zero Plot');
```



```
fprintf('Filter Order = %d\n',n);
```

Filter Order = 6

```
fprintf('Cutoff Frequency = %.4f\n',Wn);
```

Cutoff Frequency = 0.5000